

1. During gradient descent, you observe that the cost function stops decreasing after a few iterations, even though the minimum hasn't been reached. Write down three possible reasons. --3

Reason 1:

Learning rate too small → Tiny updates, progress slows.

Reason 2:

Poor feature scaling

Reason 3:

Saddle point or flat region in the cost function

2. In a housing price prediction model, you have *bedrooms*, *square_feet*, and *location*. The linear regression is underperforming. You notice that houses with the same *square_feet* and *bedrooms* are priced differently across *location*. Write down two major steps to improve performance. --4

First apply feature rescaling. Then feature engineering.

There is likely an **interaction** between location and the other features. You should consider **interaction terms**, such as:

$\text{square_feet} \times \text{location}$

3.

Consider a simple linear regression model:

$$\hat{y} = wx + b$$

Given a dataset with m examples $\{(x^{(i)}, y^{(i)})\}_{i=1}^m$, the **squared error loss** is defined as:

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m \left(\hat{y}^{(i)} - y^{(i)} \right)^2 = \frac{1}{2m} \sum_{i=1}^m \left(wx^{(i)} + b - y^{(i)} \right)^2$$

--8

a. Derive the partial derivative of the cost function $J(w,b)$ with respect to w and b .

For w:	For b:
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b. Write the gradient descent update rules for w and b using a learning rate α (alpha).

For w:	For b:
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c. Given the following data: input $X = [1, 2, 3]$, output $Y = [2, 4, 6]$, and initial parameters $w=0$, $b=0$, perform one step of gradient descent with $\alpha=0.1$.